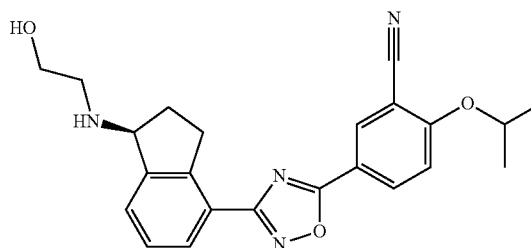


(10) **Patent No.:** US 11,111,223 B2
(45) **Date of Patent:** Sep. 7, 2021

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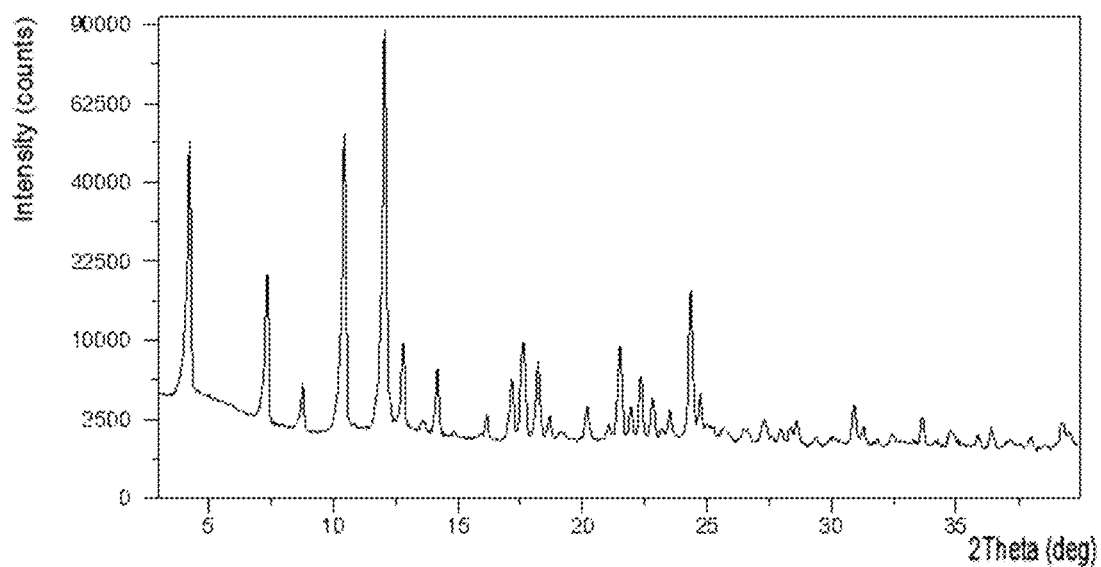


FIG.1

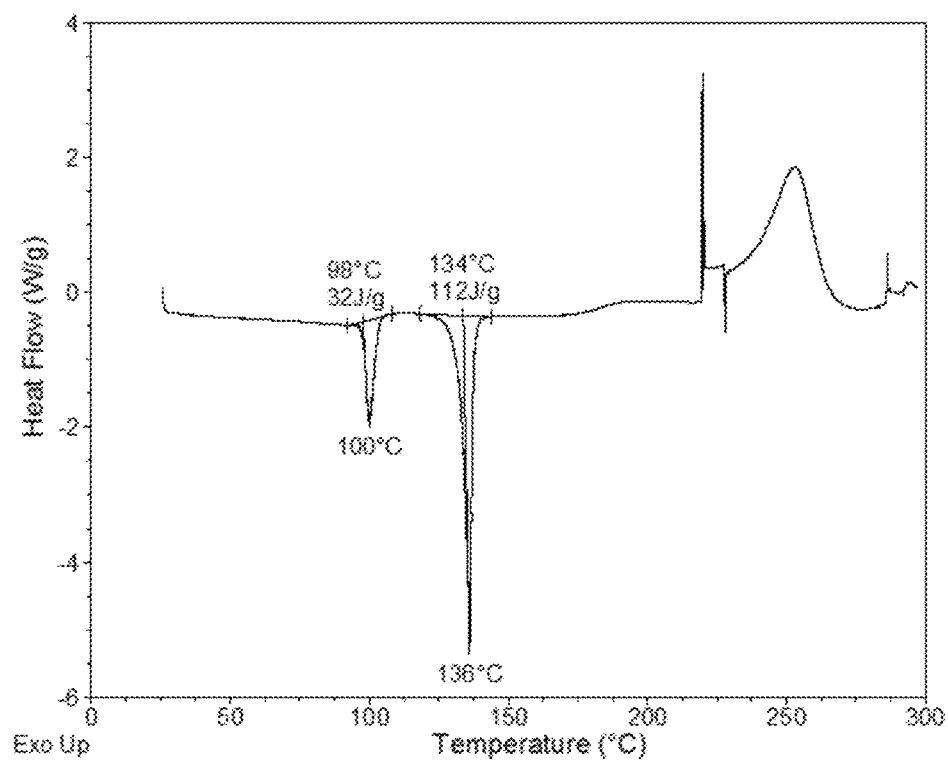


FIG.2

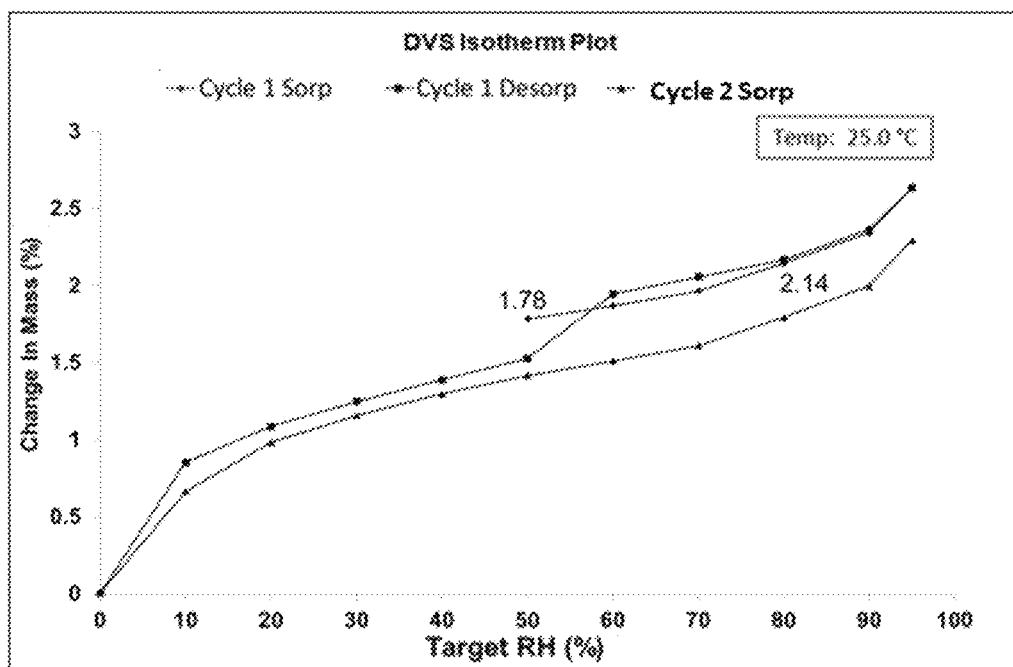


FIG.25

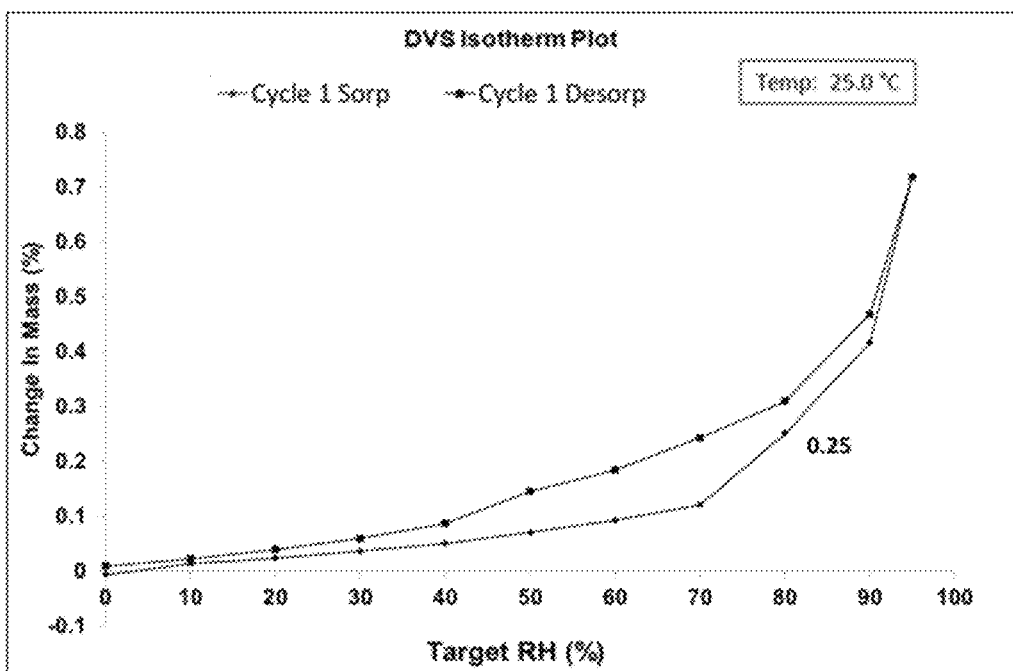


FIG.26

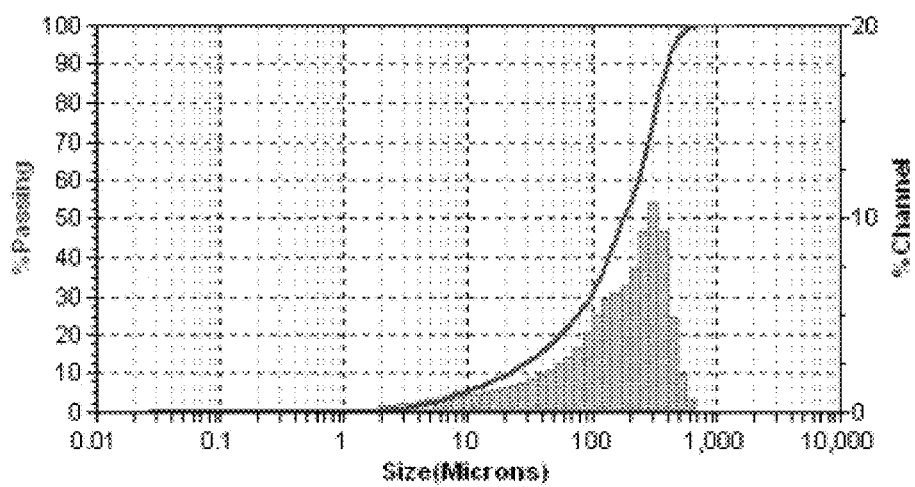


FIG.42

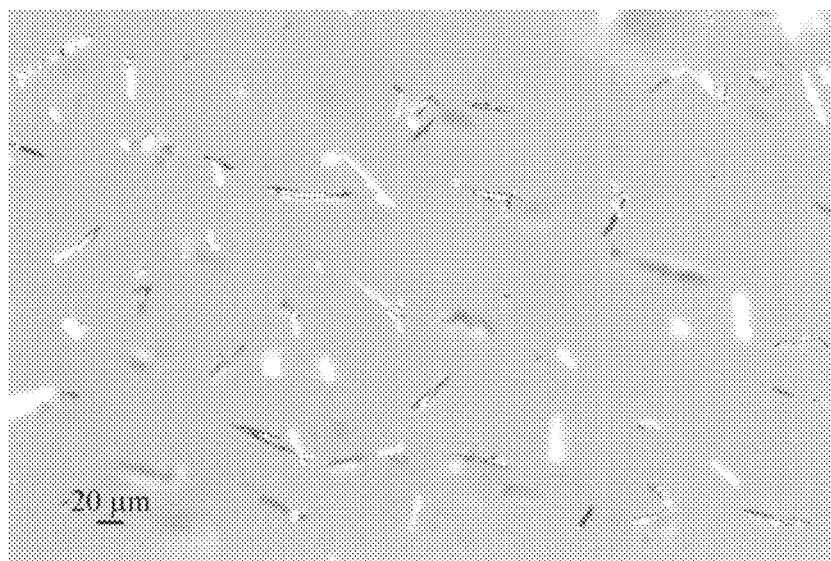


FIG.43

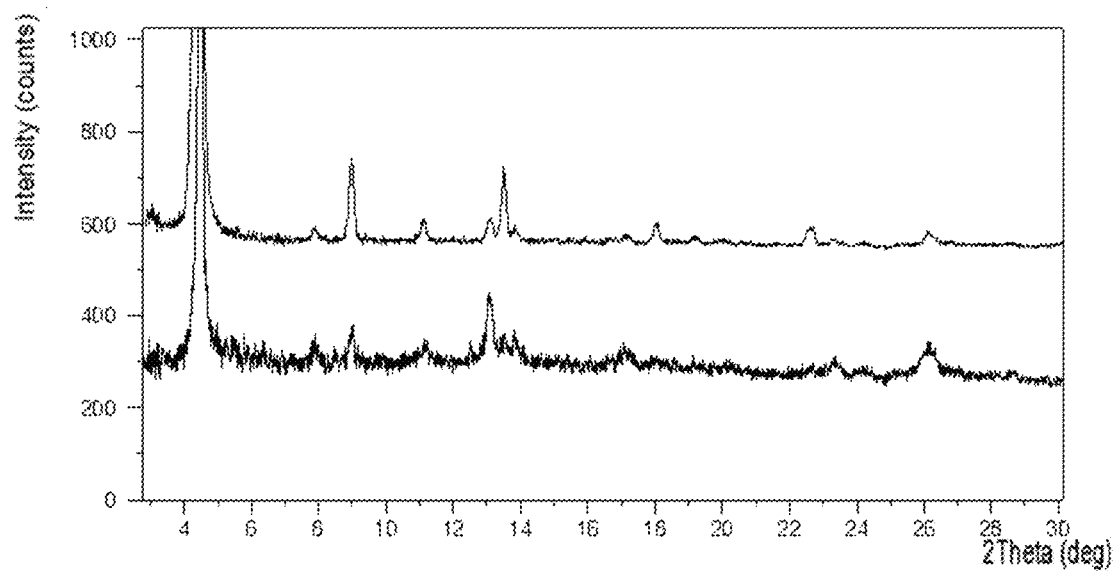


FIG.50

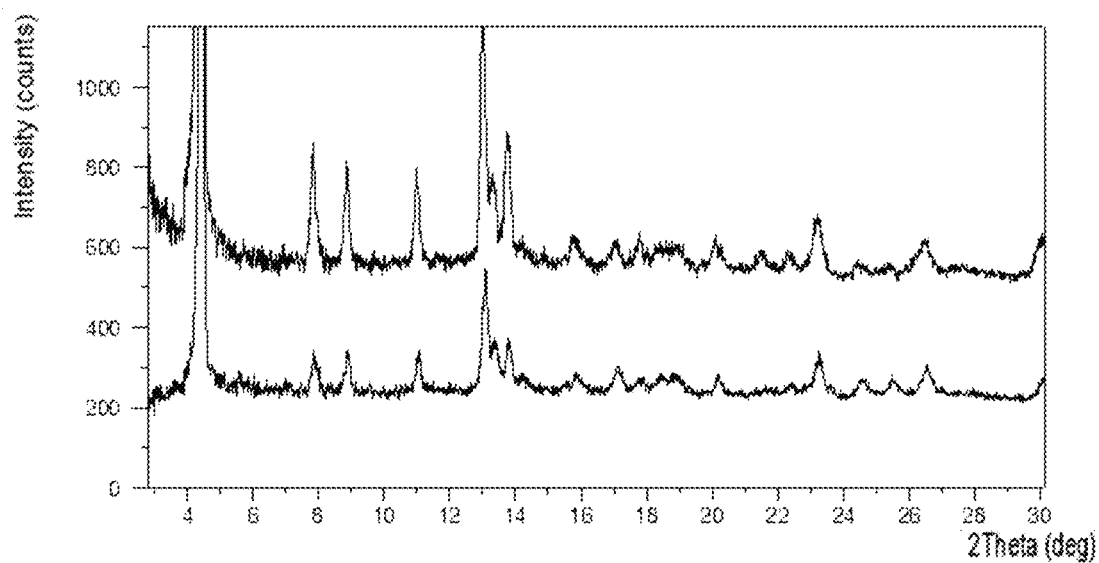


FIG.51

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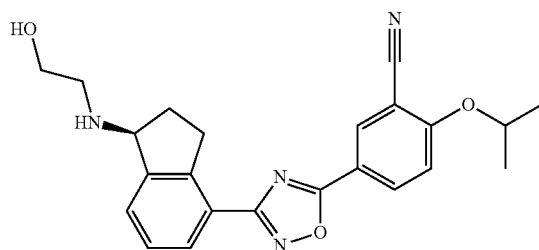
CRYSTALLINE FORMS OF OZANIMOD AND OZANIMOD HYDROCHLORIDE, AND PROCESSES FOR PREPARATION THEREOF

TECHNICAL FIELD

The present disclosure relates to the field of pharmaceutical chemistry, particularly relates to crystalline forms of ozanimod and ozanimod hydrochloride, and processes for preparation thereof.

BACKGROUND

Ozanimod is a novel, oral, selective modulator of sphingosine-1-phosphate 1 receptor (S1P1R) developed by Receptos for the treatment of autoimmune diseases. ozanimod is in phase III trials in the USA for the treatment of multiple sclerosis (MS) and ulcerative colitis (UC). Ozanimod has very excellent pharmacokinetics, pharmacodynamics and safety data in clinical trials, which can perfectly meet the differentiated development strategy and is expected to be the best second-generation S1P1R modulator drug. The chemical structure of the drug is shown as formula (I).



A specific pharmaceutical activity is the basic prerequisite to be fulfilled by a pharmaceutically active agent to be approved as a medicament on the market. However, there are a variety of additional requirements that a pharmaceutically active agent has to comply with. These requirements are based on various parameters which are connected with the nature of the active substance itself. Without being restricted, examples of these parameters are the chemical stability, solid-state stability and storage stability of the active substance under various environmental conditions, stability during production of the pharmaceutical composition and the stability of the active substance in the final drug products, etc.

The pharmaceutically active agent used for preparing the pharmaceutical compositions should be as pure as possible and its stability in long-term storage must be guaranteed under various environmental conditions. It is essential to prevent using pharmaceutical compositions which contain agent other than the actual active agent, for example decomposition products thereof. In such cases the content of active agent in the drug might be less than that specified. In addition it is important that the pharmaceutically active agent should be non-hygroscopic, stable both to degradation and subsequent solid form changes. If the pharmaceutically active agent is hygroscopic in the sense that it absorbs water (either slowly or over time), it is almost impossible to reliably formulate the pharmaceutically active agent into a drug as the amount of agent to be added to provide the same dosage will vary greatly depending upon the degree of

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hydration. Furthermore, variations in hydration or solid form can lead to changes in physicochemical properties, such as solubility or dissolution rate, which can in turn lead to inconsistent oral absorption of the patient. Preferably, therefore, a pharmaceutically active substance should be only slightly hygroscopic.

Accordingly, chemical stability, solid-state stability, "shelf life" and materials handling properties (such as ease of solubilizing the compound) of the pharmaceutically active substance are very important factors. In an ideal situation, the pharmaceutically active substance and any compositions containing it should be capable of being effectively stored over appreciable periods of time, without exhibiting a significant change in the physicochemical properties of the active substance such as its activity, moisture content, solubility, solid form, etc. Further, the pharmaceutically active substance usually needs to be processed to achieve a particle size suitable for inhalation and any crystalline form must be stable during such processing so that the properties of the final product are predictable and reliable. In short, in the production of commercially viable and pharmaceutically acceptable pharmaceutical compositions, it is desirable to provide the pharmaceutically active substance in a fully crystalline and stable form, wherever possible.

Different crystalline forms of the same solid chemical drug are significantly different in solubility and stability which can in turn affect the absorption and bioavailability of the drug products. CN102762100A or US2011172202A1 disclosed the compound of formula (I), while there is no solid form or crystalline form of ozanimod disclosed in the prior art. The prior art has neither guidance nor inspiration for finding the crystalline forms. Therefore, it is necessary to perform comprehensive polymorph screening of ozanimod to select a crystalline form which is the most suitable for drug development.

The inventors of the present disclosure have found several crystalline forms of ozanimod and a crystalline form of ozanimod hydrochloride through research process, which provides new choices for the preparation of ozanimod drug product.

SUMMARY

The present disclosure provides novel crystalline forms of ozanimod and ozanimod hydrochloride, and processes for preparation and use thereof.

One objective of the present disclosure is to provide a crystalline form of ozanimod, designated as Form CS1.

The X-ray powder diffraction pattern of Form CS1 shows characteristic peaks at 2 theta values of $12.1^{\circ} \pm 0.2^{\circ}$, $10.4^{\circ} \pm 0.2^{\circ}$ and $4.2^{\circ} \pm 0.2^{\circ}$ using $\text{CuK}\alpha$ radiation.

Furthermore, the X-ray powder diffraction pattern of Form CS1 shows one or two or three diffraction peaks at 2 theta values of $7.4^{\circ} \pm 0.2^{\circ}$, $24.3^{\circ} \pm 0.2^{\circ}$ and $17.7^{\circ} \pm 0.2^{\circ}$. Preferably, the X-ray powder diffraction pattern of Form CS1 shows three diffraction peaks at 2 theta values of $7.4^{\circ} \pm 0.2^{\circ}$, $24.3^{\circ} \pm 0.2^{\circ}$, $17.7^{\circ} \pm 0.2^{\circ}$.

Furthermore, the X-ray powder diffraction pattern of Form CS1 shows one or two or three characteristic peaks at 2 theta values of $12.8^{\circ} \pm 0.2^{\circ}$, $21.5^{\circ} \pm 0.2^{\circ}$ and $18.2^{\circ} \pm 0.2^{\circ}$. Preferably, the X-ray powder diffraction pattern of Form CS1 shows three diffraction peaks at 2 theta values of $12.8^{\circ} \pm 0.2^{\circ}$, $21.5^{\circ} \pm 0.2^{\circ}$ and $18.2^{\circ} \pm 0.2^{\circ}$.

Furthermore, the X-ray powder diffraction pattern of Form CS1 shows characteristic peaks at 2 theta values of